Reg. No. | | | | | | | | | | |

[14CE61]

SIXTH SEMESTER B.TECH EXAMINATION, JUNE – 2019

DESIGN OF STEEL STRUCTURES

Time: 3 Hours

Maximum Marks: 70

Instructions:

1. Answer any **FIVE** full Questions. Selecting at least **TWO** questions from each part.
2. Missing data may be suitably assumed.
3. IS 800: 2007 code book and Steel Tables are permitted.

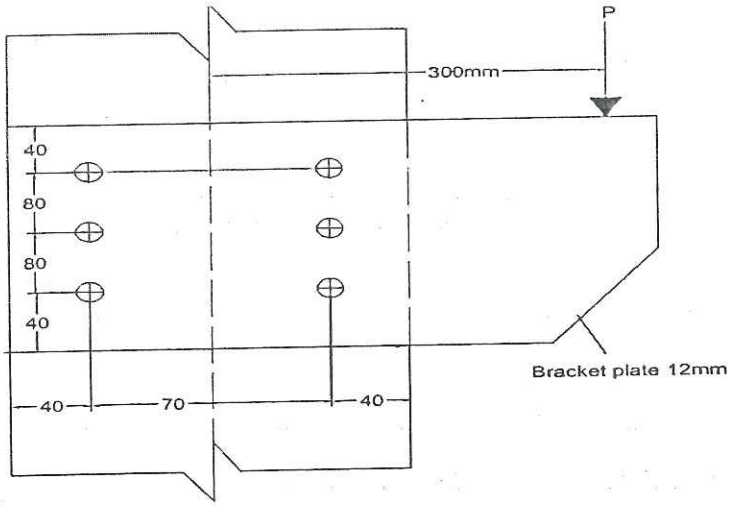
PART - A

1. a. List out the Advantages and Dis-Advantages of Steel. **06**

   b. A double cover butt joint is used to connect 2 plates of 180X20mm. 2 Cover plates of thickness 12mm is used for connection. 9 bolts of M20G4.6 are used to form each joint. Determine the strength of the joint and its efficiency. **08**

2. a. With the help of sketches explain the bolt assembly. **05**

   b. Determine the Load P that can be applied on the bracket connection shown in the figure. Use M18G4.6 bolts and Fe410. **09**



3. a. With the help of sketches explain different types of weld defects occurring in welded joints. **06**

   b. Indian Standard Angle ISA 100X75X10 is connected to the gusset plate of 12mm thick to carry an ultimate load of 200kN. Design a suitable fillet weld, if the welding is done horizontally on 2 sides and on other side vertical considering the Longer leg connected to the Gusset plate. Use Fe410 steel and shop fabrication condition. **08**

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4. A tie member of a roof truss is subjected to an ultimate tensile force of 200kN. Design a suitable unequal angle section (Single angle) whose longer leg is connected to the gusset length of the tension member between the centres of the connection is 3.2m. The member is subjected to reversal of stress due to the action of the wind. Use M16G4.6, consider the thickness of gusset as 6mm. Consider e=30mm, p=50mm and g=40mm. **14**

PART - B

5. Determine the safe load that can be applied on a steel column ISHB 400@ 806.4 N/m, the actual length of column is 5m, the ends are effectively held in position and restrained against rotation, Use Fe410 steel and determine the safe load for both X-X and Y-Y axis. **14**

6. Design a gusseted slab base for a column ISHB 400@806.4N/m to carry a working load of 2500kN. Use M20 grade concrete for foundation and Fe410 steel, thickness of gusset plate is 16mm and the angle used for connection is ISA 150X115X15 with 150mm vertical length, The connection is done with 100% bearing and the bolts used are M20G4.6, SBC of soil is 200kN/m^2. Sketch the details. **14**

7. A roof of a hall measuring 6mX13.7m consists of 120mm thick RCC slab supported on steel I-Section spaced at 3.5m centre to centre. The hall is having wall of 300mm thickness all round. The finishing load on roof is 1.0kN/m and live load is 2kN/m. Unit weight of RCC= 24 kN/m, Design the steel beam and apply all the necessary checks on design. **14**

8. Find the Ultimate Moments and Shear force for a Gantry girder using the following data: **14**

   i. Span of gantry girder= 20m
   ii. Capacity of the crane= 250kN
   iii. Self weight of the crane excluding crab= 200kN
   iv. Weight of the crab= 60kN
   v. Wheel base distance= 3.4m
   vi. Minimum hook approach= 1.1m
   vii. Self weight of rail= 0.3kN/m
   viii. Span of gantry Girder= 7m